

Final Test

- I) This is a cumulative exam.
 - II) Write your name, the date, and the course title (i.e. Trigonometry).
 - III) Show all work.
 - IV) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
 - V) You will need a calculator for this test. When π is required, use the π button on your calculator. Do not use 3.14.
 - VI) Each problem is worth eight points for a total of 200 points.
- ① Find the value of the expression below without a calculator.

$$\operatorname{arcsec} 2$$

- ② The coordinates of a point are given below. If this point lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

$(-9\sqrt{3}, -9)$



i) $\sin \theta$

ii) $\cos \theta$

iii) $\tan \theta$

iv) $\csc \theta$

v) $\sec \theta$

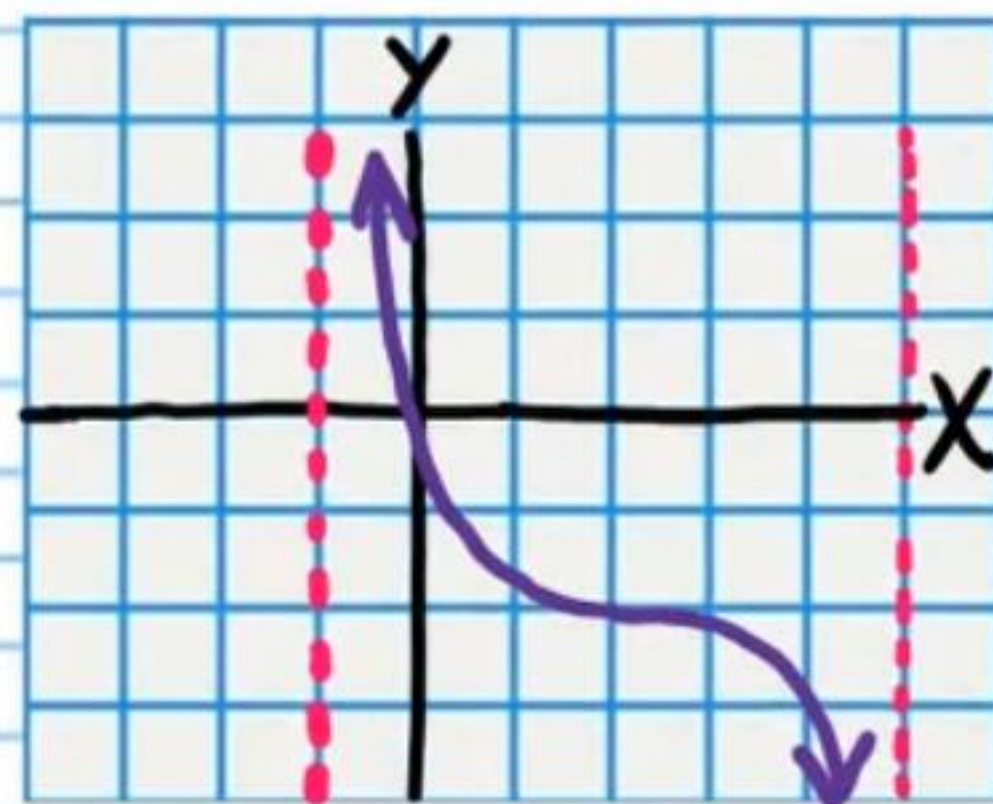
vi) $\cot \theta$

③ Write an equation corresponding to the graph below. Each horizontal unit on the graph is 90° . Each vertical unit is one.

Midline Equation:

A: ~~*~~ Assume $A=1$

Omega:



Phase Shift:

④ Find the values of the following expressions **without a calculator**.

$$\csc 15^\circ$$

⑤ i) $\cot(\cot^{-1} 1)$

ii) $\csc^{-1}(\csc(\frac{5\pi}{6}))$

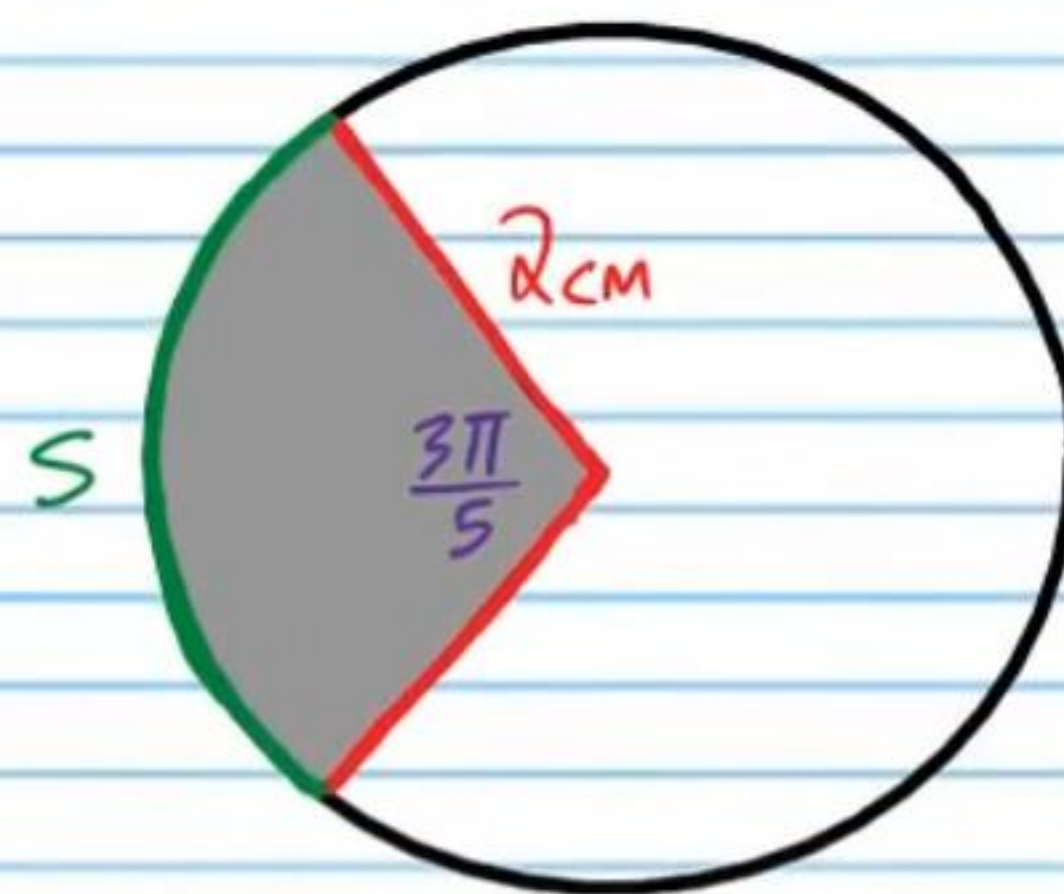
⑥ $\cos(\tan^{-1}(-\sqrt{3}))$

⑦ $\csc^{-1}(\sec(-225^\circ))$

⑧ Simplify the expression below.

$$\cos(3\theta)\cos(2\theta) + \sin(3\theta)\sin(2\theta)$$

⑨ Find the length of arc "s" on the circle.



⑩ Find the area of the shaded region in the circle.

① Find the missing side lengths and angle measures of the triangle. In other words, solve the triangle. A triangle may not be possible.

$$a = 5 \text{ in}, c = 12 \text{ in}, A = 37^\circ$$

You may or may not need this page.

⑫ Albert runs on a circular track at a constant pace. He runs 4 laps in 20 minutes. If the radius of the track is 110 feet, how many feet does he run per second? Round to the nearest thousandth.

Prove that the following equations are true by converting the left side to the right.

⑬ $\csc W (\sin W + \cos W) = 1 + \cot W$

⑭ $\sin A \cos A + \sin(2A) = 3 \sin A \cos A$

$$\textcircled{15} \frac{2\tan(\theta+180^\circ) + \tan(-\theta)}{\cos(360^\circ-\theta)} = \frac{\sin\theta}{\cos^2\theta}$$

$$\textcircled{16} \cos^2\alpha + \cot^2\alpha + \sin^2\alpha - 1 = \frac{\cos^2\alpha}{\sin^2\alpha}$$

⑪ Solve if $0 \leq A < 2\pi$. Write your answer(s) in radian units.

$$\cos(2A) = -\frac{1}{\sqrt{2}}$$

⑫ Solve if $0 < \theta < 2\pi$. Write your answer(s) in radian units.
Round to the nearest hundredth.

$$3\cot\theta + 1 = 2\cot\theta - 1$$

⑬ Find all values of β .

$$\sin\beta - 10 = -2\sin\beta - 7$$

②0 Solve if $0 < B < 360^\circ$. * Hint: You must use the quadratic formula.

$$\sec^2 B = 3 \tan B$$

②1 Solve if $0 \leq W \leq 2\pi$. Write your answer(s) in radian units.

$$2 + \sin W + \cos(2W) = 0$$

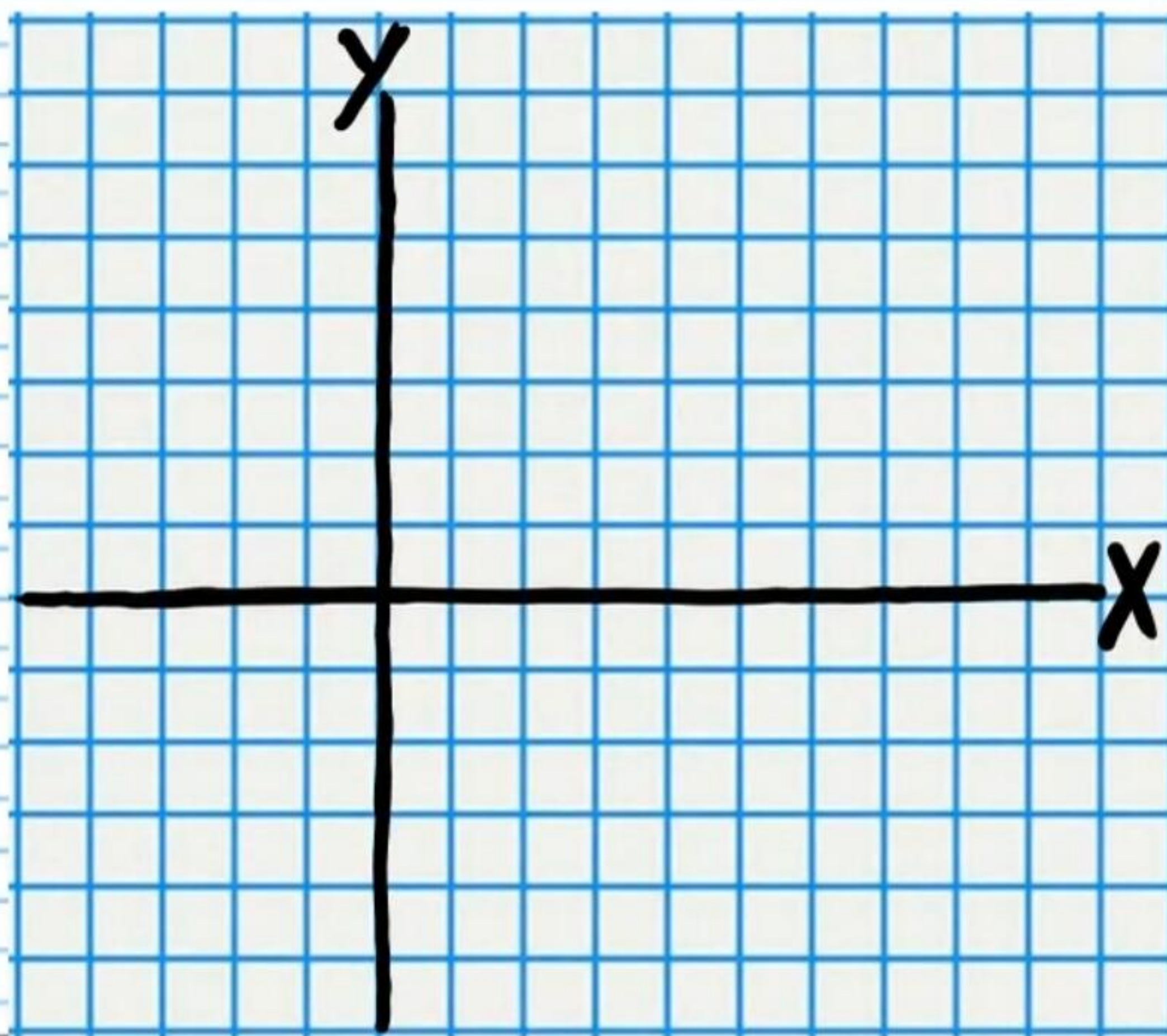
22) Graph the equation below. Write the period, phase shift, and the equation of the midline.

$$y = 2 \sec\left(\frac{x}{3} + 60^\circ\right) - 1$$

Midline Equation:

Period:

Phase Shift:

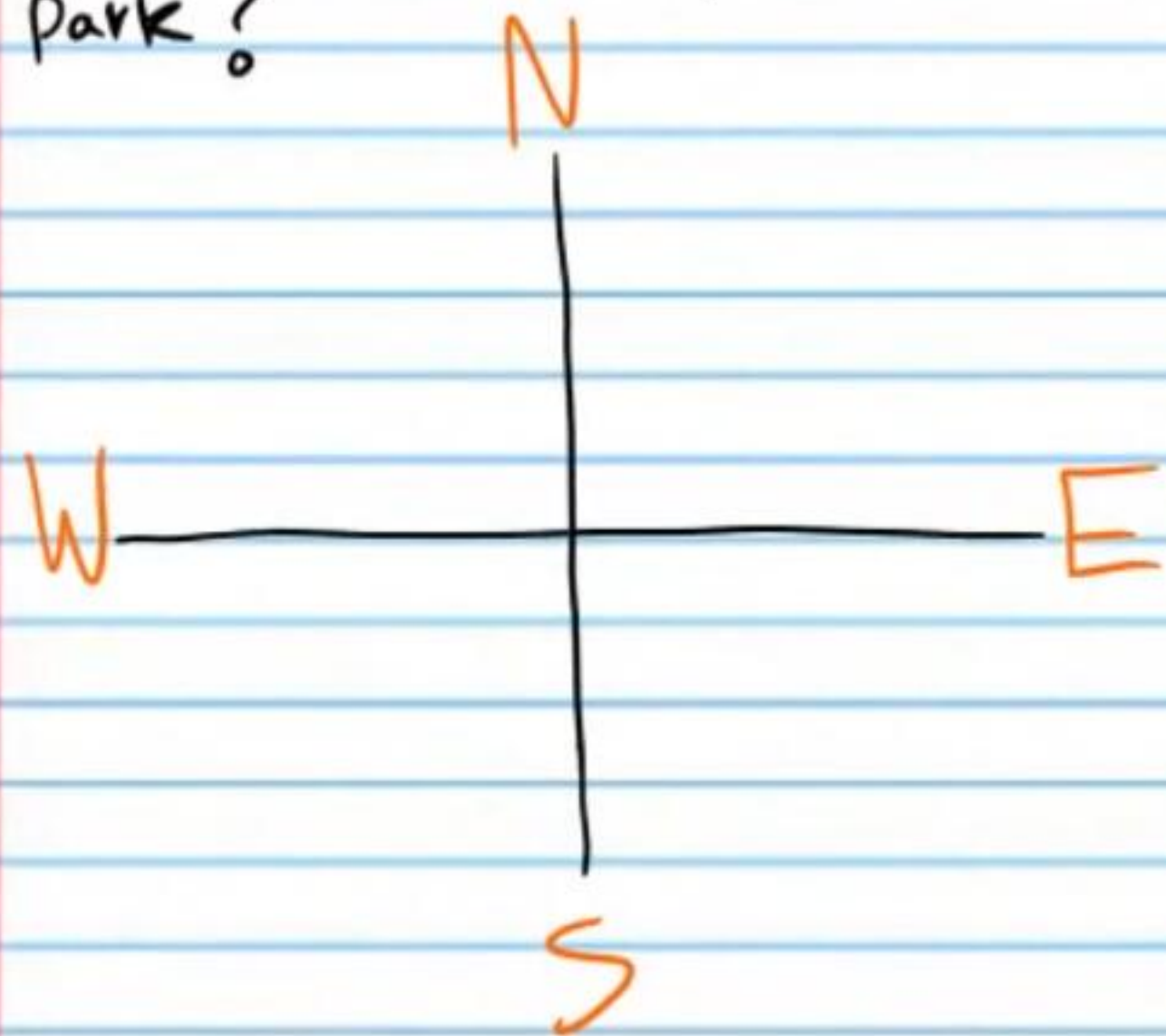


23) If α is in standard position, and it terminates in QIV ($270^\circ < \alpha < 360^\circ$), find the value of each expression below if $\cos \alpha = \frac{1}{4}$.

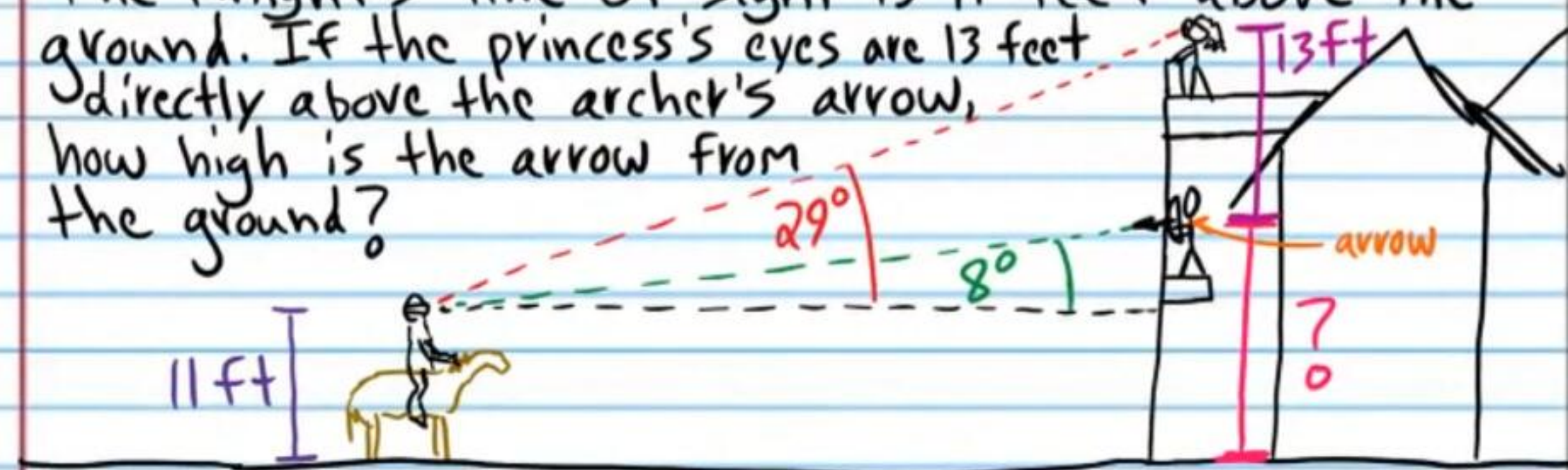
i) $\cos(2\alpha)$

ii) $\sin\left(\frac{\alpha}{2}\right)$

②4 A gps device (global positioning system) shows that a car must drive 28 miles on a heading 41 degrees west of north ($N41^\circ W$) to get to the Dublin railway. The device also shows that the car must drive 18 miles on a heading 54 degrees east of north ($N54^\circ E$) to get to the Seven Flags amusement park. How far apart are the railway and the amusement park?



②5 A knight on a horse sees a princess looking down from a castle balcony. The angle of elevation when he looks up is 29° . He then looks down at an archer in the castle with an 8° angle of elevation. The knight's line of sight is 11 feet above the ground. If the princess's eyes are 13 feet directly above the archer's arrow, how high is the arrow from the ground?



Final Test Answers

① 60°

② i) $\sin \theta = -\frac{1}{2}$

ii) $\cos \theta = -\frac{\sqrt{3}}{2}$

iii) $\tan \theta = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$

iv) $\csc \theta = -2$

v) $\sec \theta = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$

vi) $\cot \theta = \sqrt{3}$

③ $y = \cot\left(\frac{x}{3} + 30^\circ\right) - 2$

④ $\frac{4}{\sqrt{6}-\sqrt{2}}$ or $\sqrt{6} + \sqrt{2}$

⑤ i) 1 ii) $\frac{\pi}{6}$

⑥ $\frac{1}{2}$

⑦ -45°

⑧ $\cos \theta$

⑨ $\frac{6\pi}{5} \text{ cm}$

⑩ $\frac{6\pi}{5} \text{ cm}^2$

⑪ No Triangle Possible
 $\sin C \approx 1.444$

⑫ 2.304 ft/s

⑬ $\csc W(\sin W + \cos W) = 1 + \cot W$

$\csc W \sin W + \csc W \cos W =$

$\frac{1}{\sin W} \sin W + \frac{1}{\sin W} \cos W =$

$1 + \frac{\cos W}{\sin W} =$

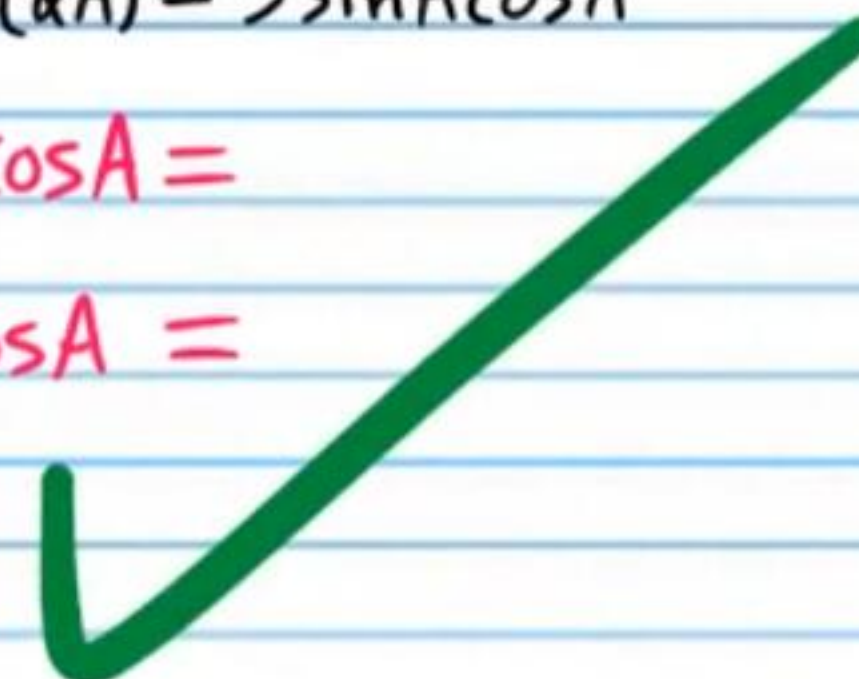
$1 + \cot W =$



⑭ $\sin A \cos A + \sin(2A) = 3 \sin A \cos A$

$\sin A \cos A + 2 \sin A \cos A =$

$3 \sin A \cos A =$



⑮ $\frac{2 \tan(\theta + 180^\circ) + \tan(-\theta)}{\cos(360^\circ - \theta)} = \frac{\sin \theta}{\cos^2 \theta}$

$\frac{2 \tan \theta - \tan \theta}{\cos \theta} =$

$\frac{\tan \theta}{\cos \theta} =$

$\frac{\frac{\sin \theta}{\cos \theta}}{\cos \theta} =$

$\frac{\sin \theta}{\cos \theta} \div \cos \theta =$

$\frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\cos \theta} =$

$\frac{\sin \theta}{\cos^2 \theta} =$



$$(16) \cos^2 \alpha + \cot^2 \alpha + \sin^2 \alpha - 1 = \frac{\cos^2 \alpha}{\sin^2 \alpha}$$

$$\sin^2 \alpha + \cos^2 \alpha + \cot^2 \alpha - 1 =$$

$$1 + \cot^2 \alpha - 1 =$$

$$\cot^2 \alpha =$$

$$\frac{\cos^2 \alpha}{\sin^2 \alpha} =$$



$$(17) A = \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{11\pi}{8}, \frac{13\pi}{8}$$

$$(18) \theta \approx 2.68, 5.82$$

$$(19) \beta = 90^\circ + 360n, n \text{ is any integer.}$$

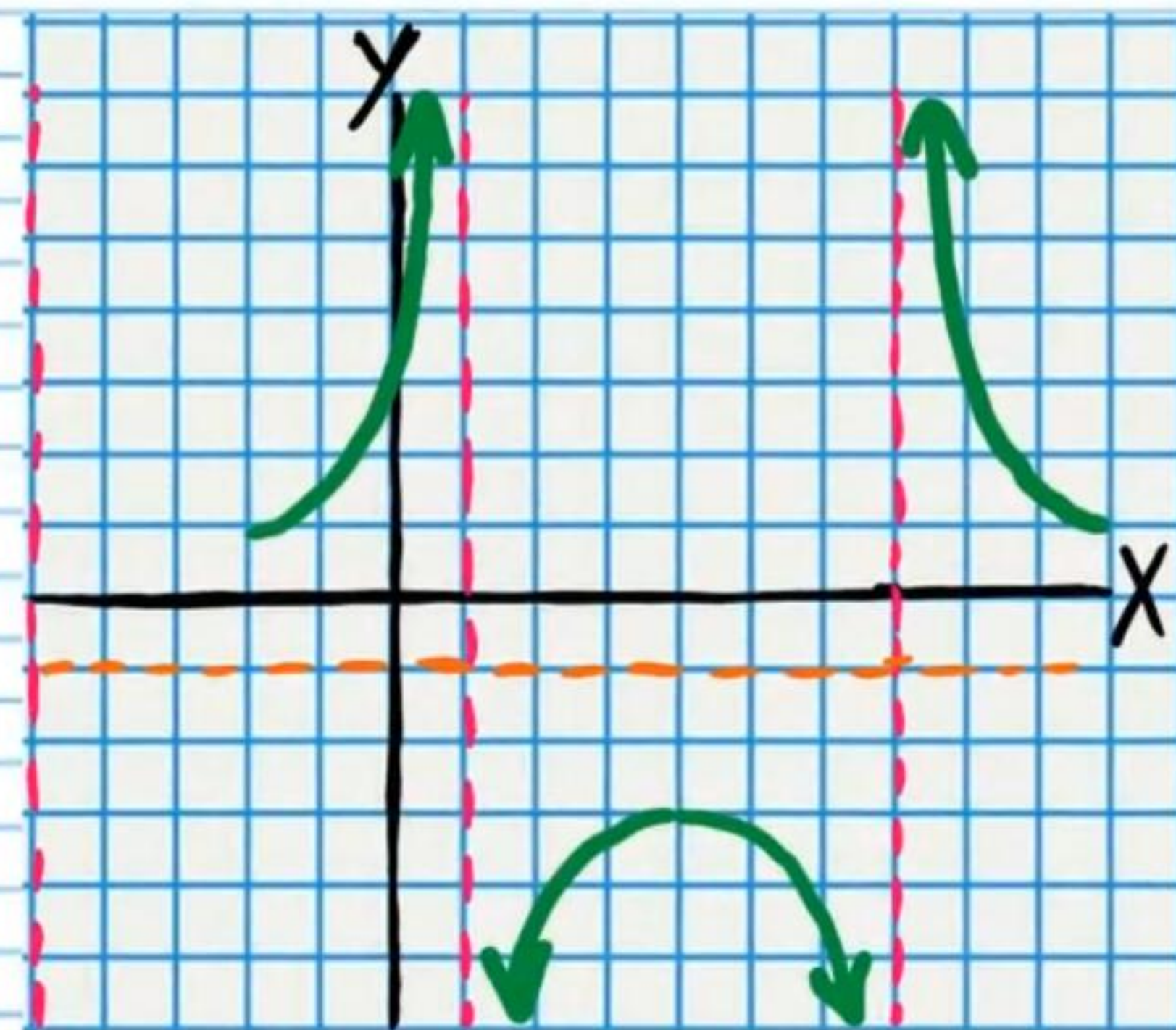
$$(20) \beta \approx 20.9^\circ, 69.09^\circ, 200.9^\circ, 249.09^\circ$$

$$(21) W = \frac{3\pi}{2}$$

$$(22) y = -1$$

$$P = 1,080^\circ$$

$$PS = -180^\circ$$



$$(23) \text{ i) } -\frac{7}{8} \quad \text{ ii) } \frac{\sqrt{6}}{4}$$

$$(24) 34.58 \text{ m}$$

$$(25) 15.416 \text{ ft}$$